

EDUCATION

Michigan State University - East Lansing, MI

Expected May 2031

Ph.D. in Electrical and Computer Engineering

Honors: Recipient of the University Distinguished Fellowship

Michigan State University - East Lansing, MI

May 2026

Bachelor of Science in Electrical Engineering

GPA: 4.0

Bachelor of Science in Computer Engineering

GPA: 4.0

Skills: C/C++, Python, MATLAB, Verilog, Bash, FreeRTOS, TensorRT, PyTorch, OpenCV, Git, Linux

Coursework: Advanced VLSI Design, Analog IC Design, Radio Frequency Electronic Circuits, Computer Systems

EXPERIENCE

Intelligent Edge Microsystems (I-EML) Lab - East Lansing, MI

Graduate Research Assistant

June 2025 - Present

Undergraduate Research Assistant

January 2025 - May 2026

- Devised a dual-branch CNN architecture that fuses visual features with explicit geometric metadata, achieving 1.46% MAPE; first-authored an IEEE manuscript detailing this low-cost alternative to depth sensors.
- Optimized deep learning inference on NVIDIA Jetson AGX Orin, utilizing TensorRT to implement network pruning and INT8 quantization, maximizing throughput under strict embedded memory and power constraints.
- Constructed a high-performance C++ augmentation engine utilizing multithreading to saturate CPU throughput, achieving a $2.3\times$ speedup over standard Python implementations to eliminate training data bottlenecks.
- Implemented a generative AI workflow using Stable Diffusion 3 to synthesize labeled training data, systematically addressing class imbalance and reducing physical data collection overhead.

DayDream Inc. - Grand Rapids, MI

Firmware Engineering Intern

August 2024 - July 2025

- Developed a low-latency sensor acquisition layer in C for the ESP32-S2, writing custom I²C drivers to interface with a BMI270 IMU and rotary encoders via high-priority hardware interrupts.
- Designed a preemptive multitasking architecture using FreeRTOS, utilizing binary semaphores and mutexes to synchronize 6 concurrent tasks across dual Xtensa LX7 cores without race conditions.
- Integrated a custom BLE GATT profile to stream high-throughput sensor telemetry to mobile clients, optimizing payload sizes and connection intervals to maximize bandwidth.
- Captured mixed-signal schematics for an AR wearable in Altium Designer, integrating switching regulators, LDOs, and noise-sensitive sensor interfaces.

PROJECTS

Two-Stage Miller-Compensated CMOS Operational Amplifier

April 2026

- Designed and verified a two-stage CMOS op-amp in SkyWater 130nm technology using Cadence Virtuoso and Spectre, achieving an open-loop DC gain of 70.17 dB and a 28.84 MHz GBW under a 1.8V supply
- Optimized a frequency compensation network and bias network to guarantee a stable 63.84° phase margin under a 10 pF load while constraining power dissipation to 789.6 μ W
- Validated high noise immunity and signal integrity across the frequency spectrum, achieving a 68.16 dB CMRR and a 91.89 dB DC PSRR.

Octapath - 8-bit microprocessor datapath

January - June 2025

- Implemented an 8-bit datapath in 0.5 μ m CMOS technology, integrating a high-speed Manchester Carry-Lookahead ALU, logarithmic barrel shifter, and dual-read/single-write 8x8 6T SRAM array.
- Produced a custom standard cell library with uniform 21 μ m pitch to enable modular hierarchical layout; performed schematic capture, physical layout, and iterative DRC/LVS verification for all functional blocks.